

Neuroscience background: from the neuron to the model neuron

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Lecture 8 opens with a real neuron and replaces it with a model neuron: a weighted sum of inputs passed through a nonlinearity. These notes give the biology behind that step, for readers without a biology course. Section 7 collects the correspondences in one table.

1. The parts of a neuron

A neuron is a cell specialized to receive, combine and send electrical signals (Figure 1).

- *Dendrites* are branching extensions that receive inputs from other neurons, at contacts called *synapses* (Section 4).
- The *soma*, or cell body, contains the nucleus. Currents from the dendrites converge on it.
- The *axon hillock* is where the axon leaves the soma; the *axon initial segment* just beyond it is packed with the channels that generate spikes (Section 2). Spikes usually start here.
- The *axon* is a thin cable, from under a millimetre to over a metre long, that carries spikes away from the soma. Its branches end in *axon terminals*, which form synapses on other neurons.

Many axons are wrapped in *myelin*, a fatty insulating sheath made by support cells (glia); the spike jumps between the gaps in the sheath, the nodes of Ranvier, and so travels much faster. Signals flow one way, from dendrites to soma to axon to the next neurons, the direction of the arrows in a network diagram.

2. The membrane and ion channels

The neuron is enclosed by a membrane, a thin layer of fat that *ions*, the charged atoms dissolved inside and outside the cell, cannot cross on their own. The ones that matter here are sodium (Na^+), potassium (K^+), chloride (Cl^-) and calcium (Ca^{2+}). Pumps keep their concentrations different on the two sides: more Na^+ outside, more K^+ inside. The voltage across the membrane, inside relative to outside, is the *membrane potential*. At rest it is about -70 mV, the *resting potential*.

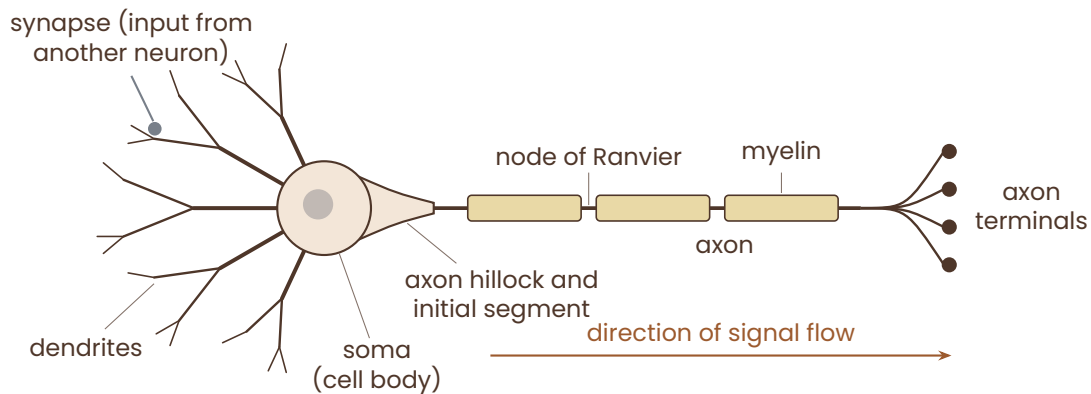


Figure 1: The parts of a neuron (schematic, not to scale).

An *ion channel* is a protein that forms a pore through the membrane. When it opens, particular ions flow through it, and this flow is an electrical current that changes the membrane potential. Na^+ flowing in makes the inside less negative (depolarization); K^+ flowing out or Cl^- flowing in makes it more negative (hyperpolarization). *Voltage-gated* channels open when the membrane potential changes; voltage-gated Na^+ and K^+ channels generate the spike. *Transmitter-gated* channels open when a chemical messenger, a neurotransmitter, binds to them; they sit at synapses and turn an incoming signal into a current.

3. Spikes and firing rates

When the membrane potential at the axon initial segment reaches a *threshold*, about -55 mV, voltage-gated Na^+ channels open. Na^+ rushes in, which opens more channels, and the potential shoots up to about $+30$ mV. The Na^+ channels then close, K^+ channels open, and K^+ flowing out brings the potential back down. This pulse is the *action potential*, or *spike*; Hodgkin and Huxley worked out its mechanism in the squid giant axon [3]. It regenerates along the axon and reaches the terminals at full size. Three properties matter for the model. A spike is *all or none*: it happens or it does not, and a neuron's spikes all have about the same size and shape, so a stronger input does not make a bigger spike. It is *brief*, about 1 ms. And it is followed by a *refractory period*: for 1–2 ms the neuron cannot fire again, and for a few milliseconds more it needs a stronger input, which caps the rate at a few hundred spikes per second. Because every spike looks the same, the usual summary of a neuron's output is its *firing rate*: spikes per second (hertz, Hz), counted over tens to hundreds of milliseconds or averaged over repeated presentations of a stimulus. Most cortical neurons fire between near zero and about 100 spikes per second.

The f - I curve

Inject a constant current I into a neuron through an electrode and count its spikes. The firing rate f as a function of I is the f - I curve. Below a threshold current the membrane never reaches spike threshold, and the rate is zero. Above it, the rate rises with the current. At large currents the rate levels off (saturates), because the refractory period sets a minimum time between spikes.

This curve is what the model neuron's nonlinearity g stands for. The rectified linear unit (ReLU), $g(z) = \max(0, z)$, keeps the first two parts: zero below threshold, rising above. The sigmoid, $g(z) = 1/(1 + e^{-z})$, keeps the rise and the saturation and smooths the threshold. Both outputs are non-negative, as firing rates are. Neither is fitted to a real cell; each keeps the threshold, which makes the output a nonlinear function of the input.

4. Synapses

A *synapse* is a contact where the axon terminal of the *presynaptic* neuron meets the *postsynaptic* neuron, across a gap of about 20 nanometres. At most synapses transmission is chemical. A spike reaching the terminal opens voltage-gated Ca^{2+} channels, and the entering Ca^{2+} makes *vesicles*, small packets of *neurotransmitter*, release their contents into the gap. The neurotransmitter binds to *receptors* on the postsynaptic membrane, many of which are transmitter-gated channels, and they open. Ions flow through them: the *postsynaptic current*. It shifts the postsynaptic membrane potential by a small amount, the *postsynaptic potential* (PSP), typically 0.1–1 mV, which fades over about ten to a few tens of milliseconds. A PSP is graded: its size varies continuously, unlike a spike. The size of the PSP that one presynaptic spike causes is the *synaptic strength*.

At an *excitatory* synapse the current depolarizes the postsynaptic neuron, toward threshold; the main excitatory transmitter in cortex is glutamate. At an *inhibitory* synapse it hyperpolarizes the neuron or holds it near rest; the main inhibitory transmitter is GABA (gamma-aminobutyric acid). By *Dale's principle*, a neuron releases the same transmitters at all of its synapses, so a cortical neuron is either excitatory or inhibitory at all of its outputs.

A typical cortical neuron receives about 10,000 synapses. Rest to threshold is about 15 mV and one excitatory PSP adds 0.1–1 mV, so tens to hundreds of inputs must arrive within the lifetime of a PSP for the neuron to fire. A neuron reports on many of its inputs being active together, not on any single one.

5. Integration

The currents from many synapses spread along the dendrites to the soma and, to a first approximation, add: two excitatory PSPs arriving together give about the sum of each alone, and an inhibitory PSP subtracts. Inputs also add over time, when one arrives before

the last PSP has faded. In terms of rates, the average current at the soma is roughly a sum over inputs of firing rate times synaptic strength, with inhibitory inputs counted negatively: the weighted sum of the model neuron.

The sum is only approximate. Dendrites have voltage-gated channels of their own and can fire local dendritic spikes, so one branch can combine its inputs nonlinearly [4]. Inhibition near the soma often opens channels that let current leak out of the cell (*shunting inhibition*), which divides the excitatory input rather than subtracting from it.

6. Plasticity: what learning changes

Synaptic strengths are not fixed, and much of learning and memory is thought to be stored in their changes. Some patterns of activity make a synapse stronger for hours, days or longer, *long-term potentiation* (LTP), first reported in the rabbit hippocampus [1]; others make it weaker, *long-term depression* (LTD). Which one happens depends on the activity of both cells, as Hebb proposed: a synapse strengthens when the presynaptic neuron repeatedly helps to fire the postsynaptic one [2]. In the model, synaptic strength is the weight w_i , and learning means changing the weights; changes in a neuron's excitability correspond roughly to the bias b . Whether the brain learns by anything like the gradient descent of this lecture is an open question.

7. From the neuron to the model neuron

The model neuron [5] keeps one number per input and one number out, and discards when the spikes happen. With D inputs, $z = \sum_{i=1}^D w_i x_i + b$ and $a = g(z)$.

Table 1: What each symbol of the model neuron stands for, and what it leaves out.

Biology	Model	Caveat
Firing rate of presynaptic neuron i	input x_i	Discards spike timing. Rates are never negative; centred inputs (mean subtracted) can be.
Synaptic strength, with its sign (excitatory +, inhibitory −)	weight w_i	A real synapse keeps its sign (Dale's principle); a model weight can change sign during learning.
Summed current at the soma	$z = \sum_i w_i x_i + b$	Only roughly additive: dendrites compute nonlinearly, and shunting inhibition divides.
Excitability: how far rest sits below threshold, plus steady background input	bias b	No single anatomical counterpart.
Spike threshold and f-I curve	nonlinearity g	ReLU or sigmoid, chosen for computation, not fitted to a cell; real f-I curves change as the neuron adapts.
Firing rate of the neuron	output $a = g(z)$	One number replaces a train of spikes; the model has no time and no noise.

8. Further reading (free online; titles are links)

- OpenStax, *Psychology 2e*, §3.2, “[Cells of the nervous system](#)”: the shortest non-technical introduction.
- OpenStax, *Biology 2e*, §35.1, “[Neurons and glial cells](#)”, and §35.2, “[How neurons communicate](#)”: more on the membrane potential, spikes and synapses.
- Neuroscience Online (UTHealth Houston), ch. 1, “[Resting potentials and action potentials](#)”, and ch. 6, “[Synaptic transmission in the central nervous system](#)”: a university-level online textbook.
- Gerstner, Kistler, Naud and Paninski, *Neuronal Dynamics*, §1.1, “[Elements of neuronal systems](#)”: the closest bridge from the biology to the model neuron.

References

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